TOLGA ARSLAN

Software Developer · Python | React | ML

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Summary

Software developer with a B.Sc. in Computer Engineering, building full-stack web applications with React, TypeScript, Node.js/Express and Python (FastAPI, Flask). Four portfolio projects, three live as public demos and backed by 700+ automated tests, span REST API design, secure authentication, SQL and NoSQL data modeling, CI, Docker deployment and ML-powered services. Internships in Agile teams.

Technical Skills

Programming & Databases: JavaScript, TypeScript, Python, SQL; PostgreSQL, MongoDB, SQL Server, SQLAlchemy

Frontend & Backend: React, Vite, HTML5, CSS3, accessible UI; Node.js, Express.js, FastAPI, Flask, REST APIs, JWT auth

Testing & DevOps: Jest, Vitest, Playwright, pytest; Git, GitHub Actions (CI), Docker, Vercel, Render

Data & ML: Pandas, NumPy, scikit-learn, LightGBM, OpenCV, PyTorch, Streamlit

Projects

Pawfect Care — Pet Health Management Platform *GitHub · Live Demo*

- Designed a 46-endpoint Express REST API with per-user data isolation, httpOnly SameSite=Strict JWT cookies, session revocation, brute-force rate limiting, CSP headers and input validation
- Wrote ~300 automated tests (170+ Jest/Supertest, 120+ Vitest/React Testing Library) run with ESLint in GitHub Actions; deployed on Vercel and Render with MongoDB Atlas

Sudoku Solver — Photo-to-Solution Web App *GitHub · Live Demo*

- Engineered a computer-vision pipeline that flattens the grid in a phone photo, reads digits with k-NN, solves the puzzle with a bitmask backtracking solver and writes the answer back onto the photo
- Solved 37 of 40 unseen newspaper photos with zero wrong solutions; photos that cannot be read reliably are flagged to the user instead of guessed

HavaUyarı — Air Quality Early Warning System *GitHub*

- Developed an end-to-end system that forecasts PM2.5 24 hours ahead for 8 stations in 5 Turkish cities and warns before unhealthy air, using Ministry SİM measurements and three Open-Meteo APIs
- Cut MAE from 12.9 to 6.8 µg/m³ in a 12-month walk-forward backtest over 62.6K station-hours: 47.7% below the raw CAMS forecast and 25.4% below the best simple baseline

Card Fraud Detection — Real-Time Transaction Scoring *GitHub · Live Demo*

- Trained a fraud model on 1.85M synthetic card transactions (Kaggle Sparkov) with point-in-time features and a pre-registered, time-based test; LightGBM reached 0.959 PR-AUC vs. 0.534 for logistic regression
- Built a FastAPI scoring service returning a decision and the top 3 SHAP reasons per transaction, with feature parity between training and serving verified on 2,000 transactions

Experience

ArVis Technology – DeepTech/AI Intern · *Teknopark İstanbul* *July 2024 – Aug 2024*

- Prepared and preprocessed training data for the CNN model of “Neurovisionist”, an AI-powered early Alzheimer’s diagnosis project, within a 16-person R&D team

VBT Yazılım A.Ş. – Software Intern *July 2023 – Aug 2023*

- Completed a 6-week training program of 50+ sessions led by VBT engineers, focused on SQL Server: joins, views, user-defined functions, stored procedures, indexing and backup/restore
- Built practice applications with C#, .NET Core and SQL Server, using Git for version control

Education

Bandırma Onyedi Eylül University *Sept 2020 – Feb 2025*

Bachelor of Science in Computer Engineering (English-taught program) *Bandırma, Türkiye*

Certifications & Training

DeepTech/AI Certificate · *ArVis, 2024* · **Data & AI Class** · *İş Bankası ProSchool* · **SQL Server Programming** · *Udemy*

Languages

Turkish: Native · **English:** B2 (Upper-Intermediate), English-taught B.Sc. program